

Kundai Farai Sachikonye

Zeltnerstrasse 30
90443, Nuernberg, Germany
☎ (+49) 1626386344
✉ kundai.sachikonye@bitspark.com
🌐 github.com/fullscreen-triangle



Data scientist: machine learning, statistics, and reproducible pipelines over large scientific datasets — from raw data to deployed, validated tool.

Profile

Computational data scientist with deep experience in high-throughput biological data — multi-omics, mass spectrometry, and imaging. I build statistically rigorous, reproducible machine-learning pipelines at scale, and take them to deployed tools validated against reference data.

Technical Skills

Analysis & statistics	Exploratory and statistical analysis; high-dimensional data; hypothesis testing; uncertainty quantification; experimental design
Machine learning	PyTorch, scikit-learn, XGBoost; CNNs; representation learning; model evaluation and benchmarking
Data engineering	Ray, Dask, SLURM/HPC; Docker, Kubernetes; HDF5, Parquet, memory-mapped I/O for >100 GB datasets; reproducible pipelines
Programming	Python (primary), R, SQL, Bash, Rust
Data domains	Genomics / transcriptomics, proteomics / metabolomics (mass spectrometry), biomedical imaging

Experience

2021–present	Independent Data Science & Computational Biology Researcher, Bitspark GmbH Full-time development of ML pipelines over multi-omics, mass-spectrometry, and imaging data; deployed, documented, reference-validated tools.
2019–2021	Computational Lipidomics — Data Analysis & HPC, Technische Universität München, Munich High-throughput analysis at HPC scale; statistical evaluation of >100 GB datasets; PyTorch classification. Taught at Master's level.
2017–2018	Glycomics & Proteomics, Universitätsklinikum Hamburg-Eppendorf, Hamburg Automated LC-MS/MS and MALDI-TOF data pipelines; reproducible workflows.
2013	Cancer Biomarker Discovery, University Medical Center Groningen, Groningen UPLC-MS/MS proteomics; analysis of high-dimensional feature sets.

Selected Projects

gospel	Multi-omics integration (genotype + transcriptomics + metagenomics); validated against 1000 Genomes, gnomAD, UK Biobank. github.com/fullscreen-triangle/gospel
lavoisier	High-throughput mass-spectrometry pipeline; 98.9% NIST accuracy; distributed (Ray). lavoisier.vercel.app/sandbox

syndrome Computational oncology — neoantigen / MHC prediction (AUC 0.81, IEDB). syndrome-seven.vercel.app

Education

2018–2019 **MSc Computational Biology**, *Universität Tübingen*

2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*

2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*

Additional: doctoral research, computational lipidomics, TUM (not completed).

Languages

English Native / Fluent (C2)

German Conversational

resident in Germany since 2011